

# Week 5 Worksheet 2

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**Exercise 1.** Denote the eigenspinors of  $S_z = \frac{\hbar}{2}\sigma_z$  for a spin-1/2 particle by  $|+\rangle$  and  $|-\rangle$ , with eigenvalues  $\hbar/2$  and  $-\hbar/2$ , respectively.

- a) Write down  $S_y = \frac{\hbar}{2}\sigma_y$  and  $S_x = \frac{\hbar}{2}\sigma_x$  as

$$a|+\rangle\langle+| + b|+\rangle\langle-| + c|-\rangle\langle+| + d|-\rangle\langle-|;$$

that is, determine  $a, b, c, d$ .

- b) Explain how you could extract  $a, b, c, d$  for  $S_i$  by taking  $\langle\pm|S_i|\pm\rangle$  or  $\langle\pm|S_i|\mp\rangle$ .

**Exercise 2. Adapted from Griffiths 4.33** Suppose we instead wanted to compute spin for a spin-1/2 particle along an arbitrary direction  $\hat{n}$  (the hats in this problem denote “unit vector,” not “operator”).

- a) Convince yourself that we can place the  $z$ -axis along  $\hat{n}$  by rotating first around  $\hat{y}$  by an angle  $\beta$  and then around  $\hat{z}$  (the old  $z$ -axis, not the new one!) by an angle  $\alpha$ .

*Hint:* Write  $\hat{n}$  in terms of its polar and azimuthal angles. How are these related to  $\alpha$  and  $\beta$ ?

- b) Use this to compute the eigenspinor of  $S_n$  with eigenvalue  $\hbar/2$ ,  $|\hat{n}; +\rangle$ : It is given by

$$|\hat{n}; +\rangle = e^{-\frac{i}{2}\alpha\sigma_z} e^{-\frac{i}{2}\beta\sigma_y} |+\rangle,$$

where  $|+\rangle$  is the eigenvector of  $S_z$  with eigenvalue  $\hbar/2$ .

*Hint:* You can use the formula which you will prove on the review:

$$e^{-i\theta\hat{n}\cdot\boldsymbol{\sigma}} = \mathbb{1} \cos \theta - i(\hat{n} \cdot \boldsymbol{\sigma}) \sin \theta.$$

- c) Check that the spinor  $|\hat{n}; +\rangle$  is an eigenspinor of  $\hat{n} \cdot \boldsymbol{\sigma}$ , which shows it is indeed an eigenspinor of  $S_n = \hat{n} \cdot \mathbf{S}$ .

*Hints:* Use the following three facts you will prove on the review: If  $H$  is hermitian, then  $e^{iH}$  is unitary. If  $U_1, U_2$  are two unitary operators, then  $U_1 U_2$  is also unitary. If  $U = e^{-\frac{i}{2}\alpha\sigma_z} e^{-\frac{i}{2}\beta\sigma_y}$ , then

$$U^\dagger \boldsymbol{\sigma} U = R \boldsymbol{\sigma},$$

where  $R$  is the  $(3 \times 3)$ -matrix which rotates the  $z$ -axis to  $\hat{n}$ . Also note that if  $\mathbf{a}, \mathbf{b}$  are 3-vectors and  $R$  is an invertible  $(3 \times 3)$ -matrix, then  $\mathbf{a} \cdot R\mathbf{b} = R^{-1}\mathbf{a} \cdot \mathbf{b}$ .

- d) Repeat (b) and (c) for  $|\hat{n}; -\rangle$ .